MIT635
Data Wrangling and Visualization
Professional Group Assignment Report
Southern Cross Logistics

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Unit	MIT635 — Data Wrangling and Visualization
Assessment	Group Assignment — Practical Work and Presentation

Report focus: workflow from raw logistics data to cleaned Python analysis and an interactive Power BI dashboard.

1. Introduction and Problem Statement

Southern Cross Logistics needs reliable data to understand its shipments, delivery performance, costs and daily operations. The provided files had useful information about shipments, customers, routes, warehouses, carriers, vehicles and delivery performance. However, the raw data was not ready to analyse. We found missing values, duplicate rows, different date formats, invalid dates, inconsistent category names, missing foreign keys and some unrealistic numbers. These problems had to be handled first because they could affect the final results.

In this project, we worked with the raw logistics data and prepared it for analysis. First, we explored and cleaned the data in Python. After that, we created charts to look at shipment volume, delivery reliability, delays, cost, profit and customer results. The cleaned files were then imported into Power BI. We connected the tables in a data model and built an interactive dashboard.

The analysis focused on a few main business questions. We wanted to know which services handle the most shipments and which ones have lower on-time performance. We also checked what is related to shipment cost, which shipment types have more delay, and whether complaints are connected with lower customer satisfaction. The dashboard was made to show these results clearly and help managers see which areas may need attention.

2. Workflow Used in the Project

We completed the work in five main stages. We started by loading and checking all seven datasets. Then we found the data-quality problems and cleaned them in Python. After cleaning, we checked the data again to make sure the main issues were fixed. Next, we created charts in Python and calculated the main performance results. Finally, we exported the cleaned CSV files to Power BI and created the relationships, DAX measures, filters, slicers and report pages. We followed this order so the Power BI dashboard was based on cleaned data instead of the original files.

Stage	Main work	Output
1. Explore	Load seven CSV tables; check shape, types, missing values, duplicates and unusual values	Data-quality profile
2. Clean	Correct dates, numeric placeholders, categories, IDs and duplicates	Clean working tables
3. Validate	Recheck duplicates, missing values, invalid numeric values and categories	Final cleaning summary
4. Analyse	Create Python charts and KPI calculations	Business findings
5. Dashboard	Export clean CSVs; build Power BI model, DAX and interactive pages	Final .pbix report

3. Data Wrangling in Python

3.1 Initial exploration

We used pandas to load the Shipments, Customers, Warehouses, Carriers, Routes, Vehicles and Delivery Performance files. At the beginning, we checked the number of rows and columns, data types, missing

values and duplicate rows. The original Shipments file had 25,750 rows, while Customers had 4,120 rows. We also noticed problems in some important fields, mainly dates, distance, weight, total cost, service names and foreign-key IDs.

We did not remove every unusual value straight away. Before deleting anything, we checked if the value could be corrected by using information from another table. This allowed us to keep useful records when possible instead of making guesses.

3.2 Delay calculation

We checked DelayDays by comparing it with PlannedTransitDays and ActualTransitDays. Some DelayDays values did not match these fields, and a few values were unusually high. Because of this, we calculated DelayDays again using ActualTransitDays minus PlannedTransitDays in both the Shipments and Delivery Performance tables. If the result was negative, we treated it as an early delivery, not as an error. This gave us a more sensible delay value for the charts and KPI calculations.

3.3 Numeric cleaning and recovery

The shipment table had some negative values in WeightKg, DistanceKm and TotalCost. It also had placeholder values such as 999999 for weight or cost and 99999 for distance. We first changed these incorrect values to missing values. For DistanceKm, we used RouteID to get the correct distance from the Routes table. This fixed 240 missing or invalid distance values. Only five were still missing because their RouteID was Unknown. For missing WeightKg, we used the median weight. This meant we could keep those shipment records in the analysis.

We also checked TotalCost using its separate cost parts. For valid records, TotalCost matched FreightCost + FuelSurcharge + InsuranceCost, apart from small rounding differences. We therefore used these parts to fix 267 missing or invalid TotalCost values. After this cleaning step, TotalCost had no missing or negative values.

3.4 Duplicates, dates and missing information

We removed exact duplicate rows from Shipments and Customers. After this, the Shipments table had 24,900 rows and Customers had 4,000 rows. We checked all seven datasets again and found no exact duplicate rows.

The dates had a few different problems. Some were written in different formats, while others were not real dates. For example, the data included 2024-02-31, 31/13/2024 and invalid_date. We converted the dates where possible. If a date was clearly wrong and we could not know the correct date, we left it as missing instead of guessing. After cleaning, Shipments still had 96 missing DispatchDate values and 101 missing DeliveryDate values. Customers had 17 missing AccountStartDate values. We kept these rows because the other information was still useful, but a chart that needs a valid date will not include them.

3.5 Categories, IDs and currency

We cleaned the category names so the same service would not appear as different groups. For example, Cold Chian, Coldchain and Cold-chain were all changed to Cold Chain. We also corrected Standrd, Expreess, Internatonal and SameDay to Standard, Express, International and Same Day. State names were cleaned in the same way and changed to NSW, VIC, QLD, WA and SA.

When a CustomerID or RouteID did not have a reliable match, we changed it to Unknown. We preferred this instead of connecting a shipment to the wrong customer or route. We also cleaned the currency names. A$ became AUD and US$ became USD. We kept a plain $ as it was because there was not enough information to know whether it meant AUD or USD. Because the dataset has different currencies, we used a currency slicer in Power BI rather than adding all currency values together.

3.6 Final cleaning check

After cleaning, we checked the data one more time. The final shipment table had 24,900 records. There were no duplicate shipments or customers, and we did not find any negative values in WeightKg, DistanceKm

or TotalCost. The known placeholder values were also gone. Five DistanceKm values were still missing because their RouteID was Unknown. We kept these records since there was no reliable distance to use. At the end, all seven cleaned tables were exported as CSV files and used in Power BI.

4. Python Visualisation and Business Findings

We used Python for more than data cleaning. Before making the Power BI dashboard, we created several charts to understand the data and the business results. We used different chart types depending on what we wanted to compare or show.

4.1 Monthly shipment volume

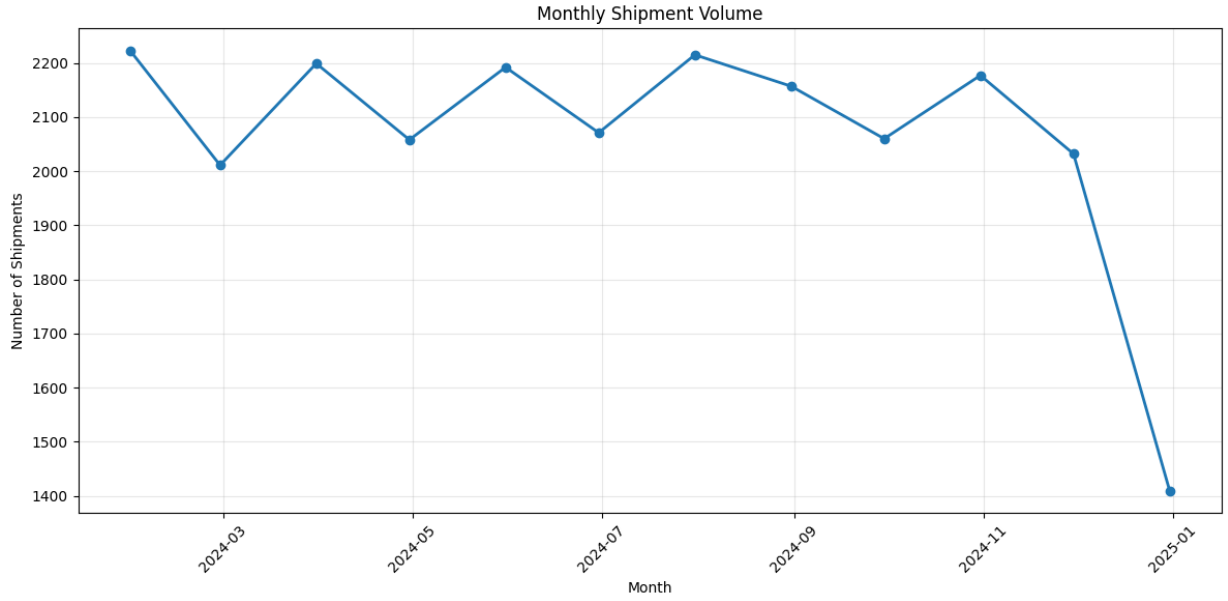


Figure 1. Monthly Shipment Volume generated in Python.

We used a line chart for monthly shipment volume because it makes the changes over time easy to see. The chart shows how shipment numbers changed during 2024. Managers can use this type of information to understand busy and quiet periods and plan staff, vehicles and warehouse capacity. We later added the same monthly trend to the Power BI Executive Overview, where users can filter the result.

4.2 Shipment volume by service level

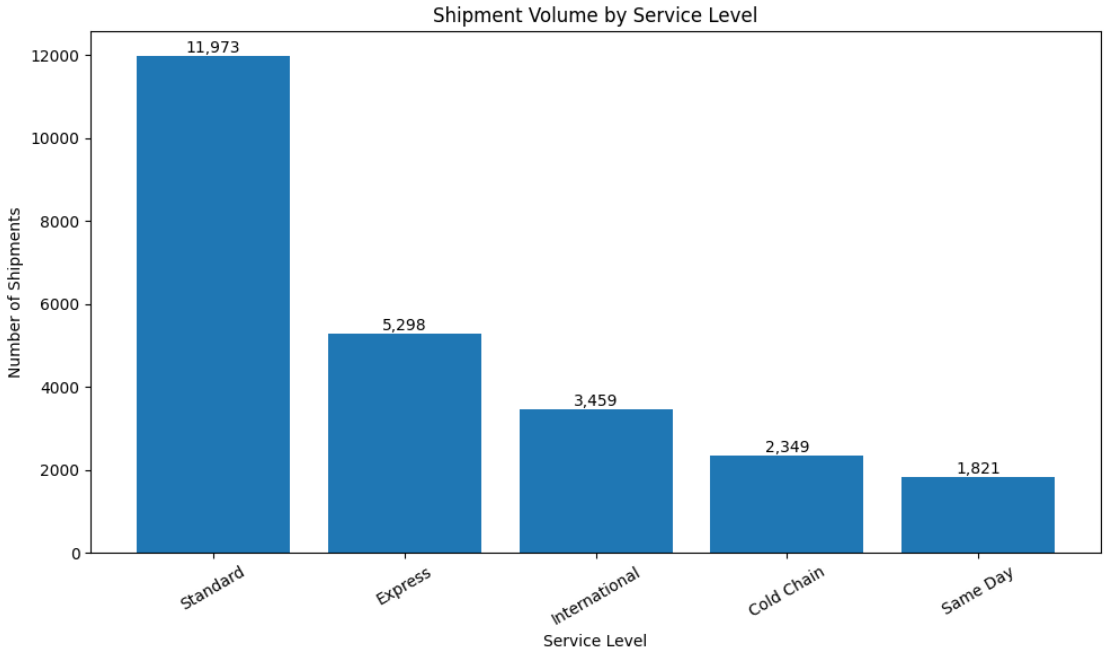


Figure 2. Shipment Volume by Service Level.

We used a bar chart to compare shipment numbers for each service level. Standard had the highest volume with 11,973 shipments. Express had 5,298 shipments, International had 3,459, Cold Chain had 2,349 and Same Day had 1,821. This shows that Standard handles a large part of the company's shipments. Because its volume is high, even a small change in Standard performance can affect many deliveries.

4.3 Distance and total cost

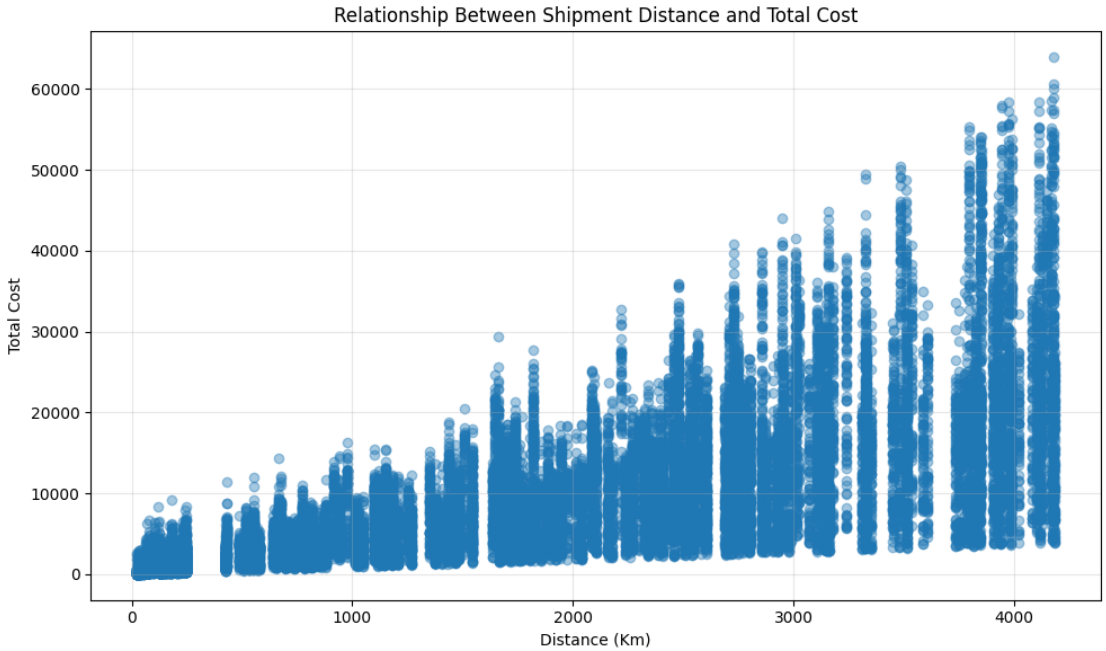


Figure 3. Relationship between Shipment Distance and Total Cost.

We used a scatter plot to see how shipment distance and total cost are related. The correlation between DistanceKm and TotalCost was 0.738. This is a fairly strong positive relationship, so longer shipments generally have a higher cost. However, distance is not the only factor. Service level, fuel surcharge,

insurance and other costs can also change the final amount. The data also contains different currencies, so cost comparisons should use the same currency or a proper currency conversion.

4.4 Cost distribution by service level

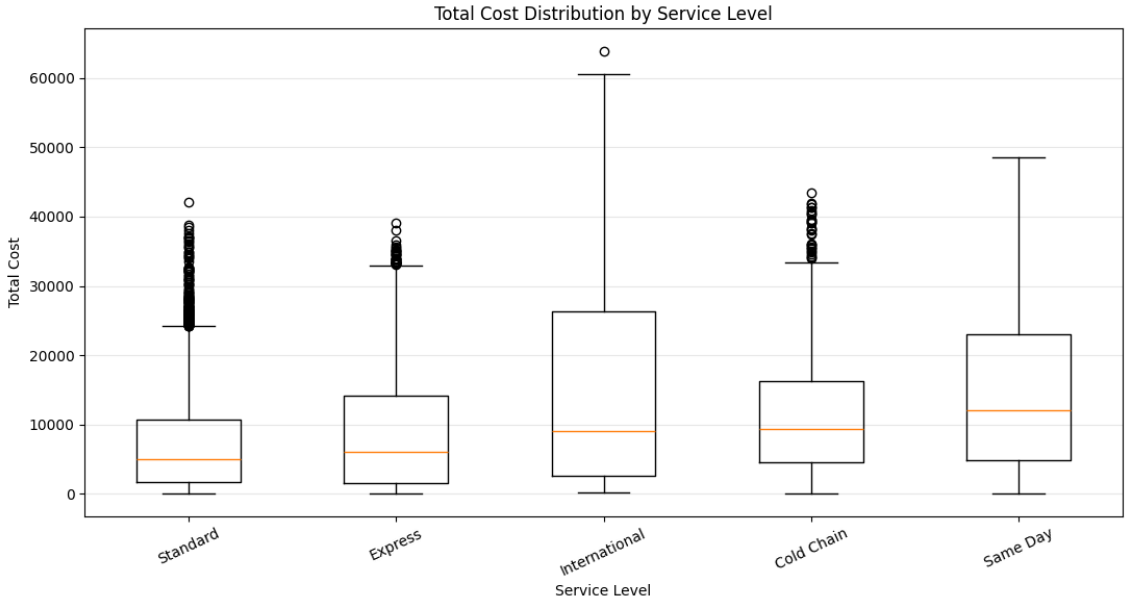


Figure 4. Total Cost Distribution by Service Level.

We used a box plot to compare costs across the service levels. It helped us see the middle cost, how much the values varied, and the unusually expensive shipments. Same Day had the highest median TotalCost at about 12,090.32. Cold Chain was around 9,323.70 and International was around 9,078.29. Express and Standard had lower median costs. International was interesting because its average cost was about 15,590.52, which was much higher than its median. This means some very expensive International shipments increased the average. These costs should still be compared using a consistent currency.

4.5 Correlation heatmap



Figure 5. Correlation Heatmap of Shipment Variables.

The correlation heatmap allowed us to check several numeric relationships together. TotalCost had a very strong relationship with FreightCost (0.998), ChargedAmount (0.994) and FuelSurcharge (0.908). It was also strongly related to Profit (0.892), InsuranceCost (0.793) and DistanceKm (0.738). The strong link with FreightCost makes sense because FreightCost is included in TotalCost. DelayDays had almost no linear relationship with TotalCost. Based on this result, delay looks more like a delivery-performance issue than a main reason for higher cost.

4.6 On-time delivery

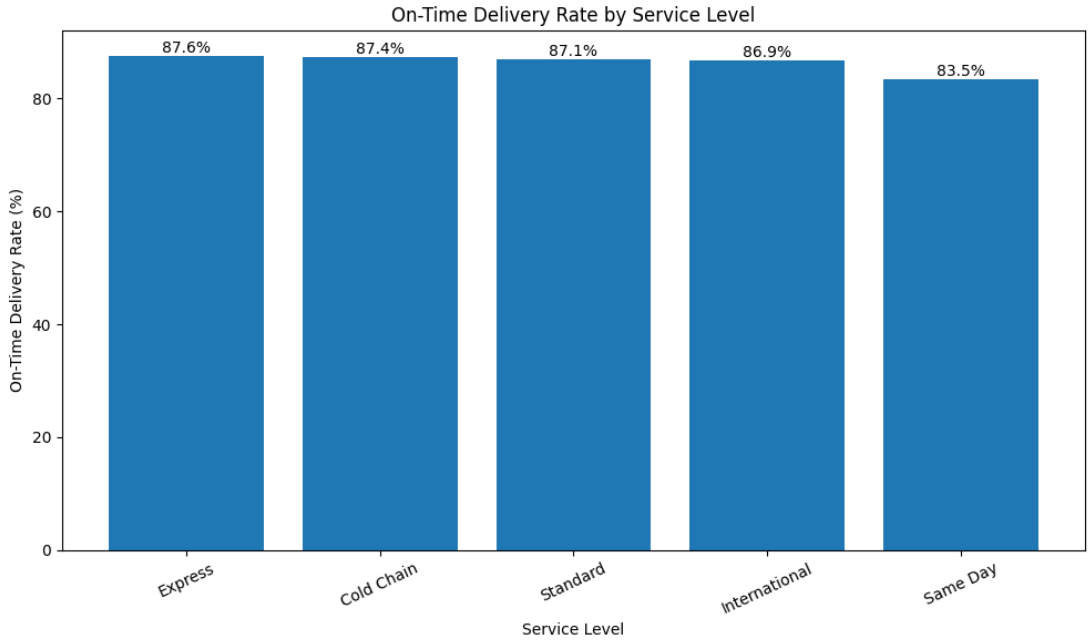


Figure 6. On-Time Delivery Rate by Service Level.

The overall on-time delivery rate was 86.93%. Express had the best result at 87.64%. Cold Chain was 87.44%, Standard was 87.05% and International was 86.87%. Same Day had the lowest rate at 83.53%. This is worth looking at because customers choose Same Day when they need a fast delivery. A lower on-time rate in this service can therefore be a bigger problem for customer expectations.

4.7 Complaints, satisfaction and delay

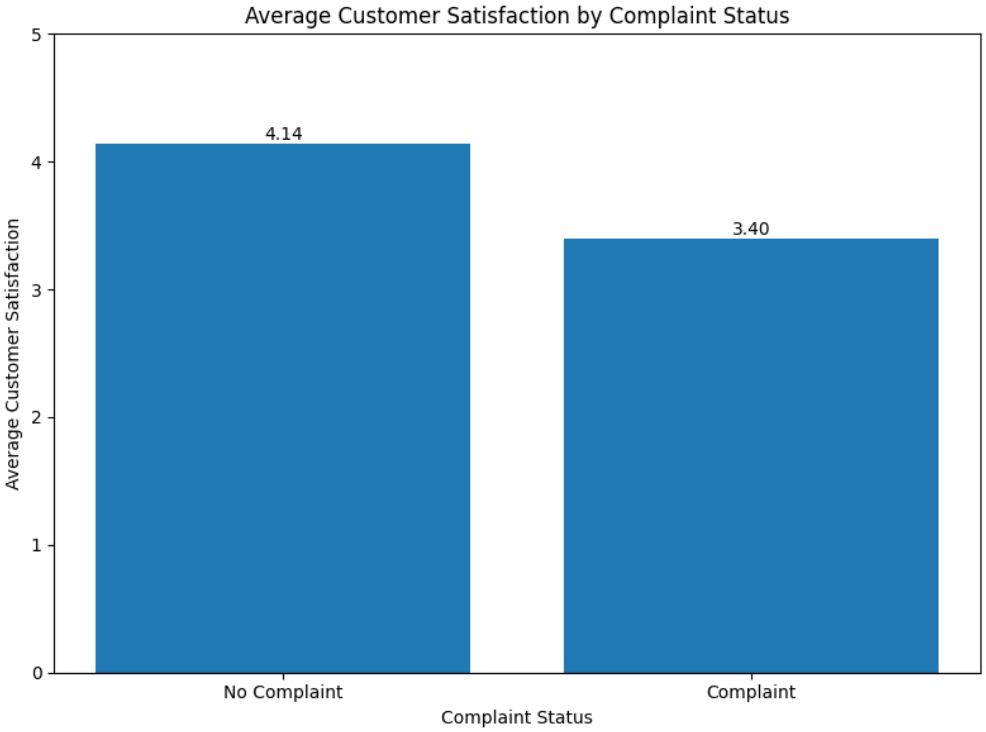


Figure 7. Average Customer Satisfaction by Complaint Status.

The data showed lower customer satisfaction when there was a complaint. Deliveries with no complaint had an average satisfaction score of 4.14, compared with 3.40 when a complaint was recorded. This does not

mean the complaint definitely caused the lower score in every case. However, it shows that complaint cases are linked with a poorer customer experience and are worth reviewing.

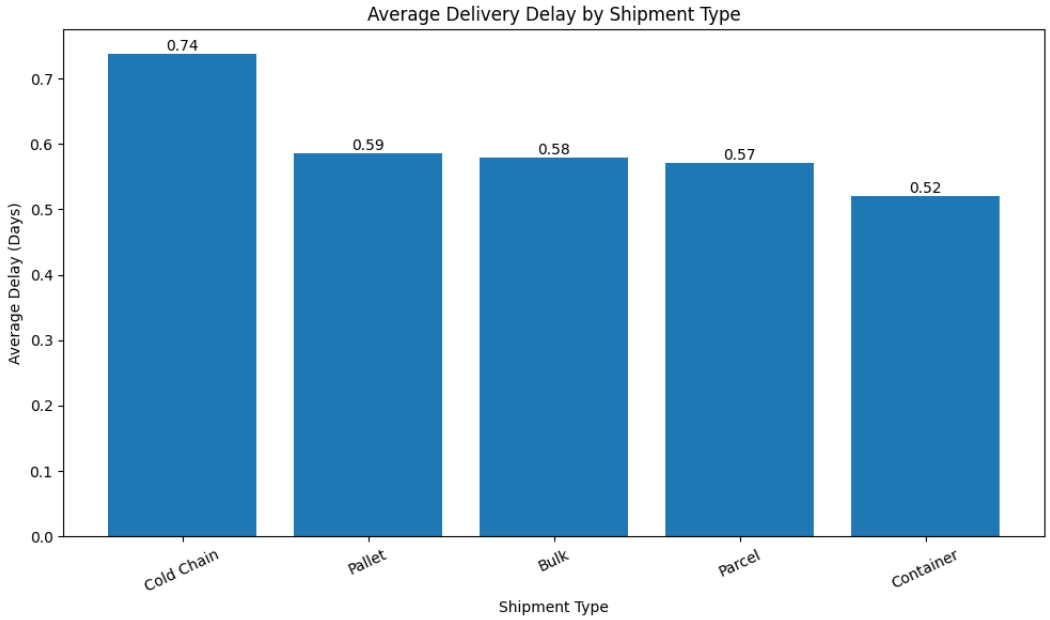


Figure 8. Average Delivery Delay by Shipment Type.

Cold Chain had the highest average delay at about 0.74 days. Pallet, Bulk and Parcel were close to the overall average, while Container had a lower average delay of about 0.52 days. Cold Chain deliveries can require special handling and stricter timing. The company could therefore check whether handling time, vehicle availability or route planning is causing some of these delays.

5. Power BI Dashboard Workflow

5.1 Data model and measures

We imported all seven cleaned CSV files into Power BI. Shipments_Clean was the main table in the model. We connected Customers, Routes, Carriers and Vehicles to it by using their ID fields. Warehouses was connected with Routes through WarehouseID and OriginWarehouseID. Delivery_Performance_Clean was linked with Shipments_Clean using ShipmentID. These relationships allowed us to use information from different tables in the same report.

We created DAX measures for the main KPIs so the values would be calculated from the data. Total Shipments counts the shipment rows. On Time Rate finds the percentage where OnTimeFlag = Y. Average Delay calculates the average DelayDays, and Damage Rate finds the percentage where DamageFlag = Y. The dashboard results were about 24.9K shipments, an 86.93% on-time rate, 0.59 average delay days and a 3.03% damage rate.

5.2 Dashboard pages and interactivity

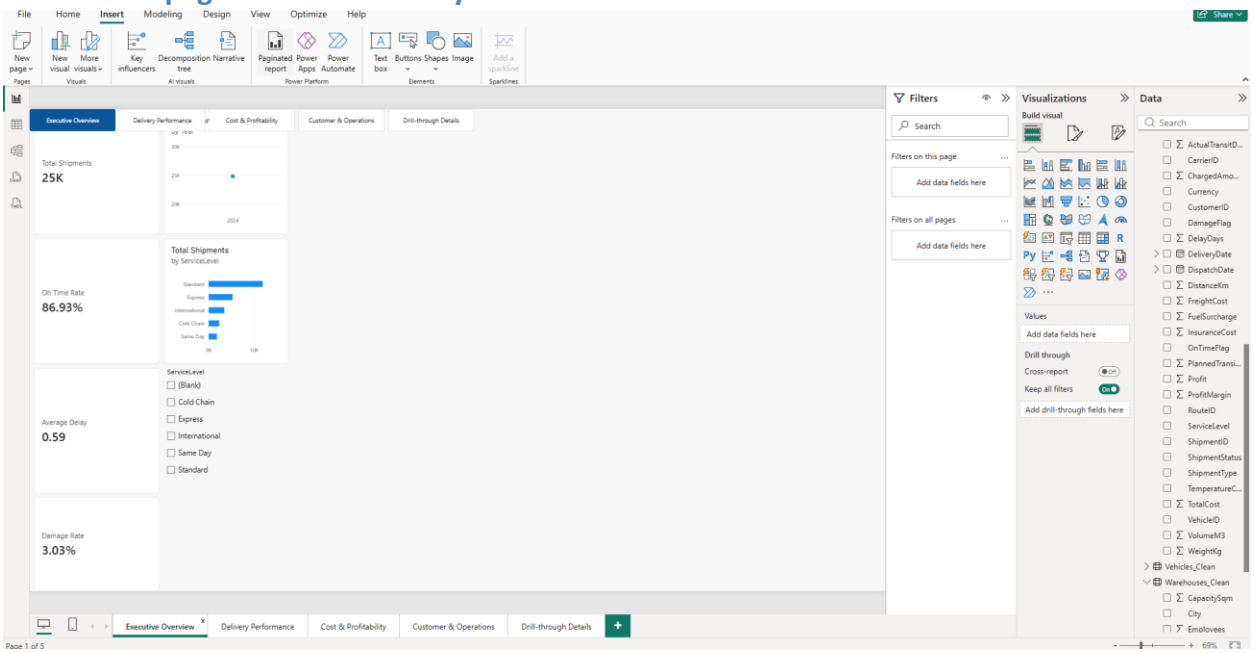


Figure 9. Power BI Executive Overview with KPI cards, monthly trend, service-level analysis, slicer and page navigation.

The Executive Overview page shows the main results in one place. It has KPI cards for Total Shipments, On Time Rate, Average Delay and Damage Rate. It also shows the monthly shipment trend, shipment volume by Service Level and a Service Level slicer. We added navigation buttons at the top so users can move between the report pages more easily.

The Delivery Performance page is mainly about on-time delivery, delay and damage. The Cost & Profitability page compares cost and profit for the different service levels. It also has a Currency slicer. When we reviewed the money-related charts, we selected AUD so different currencies were not added together by mistake. The Customer & Operations page shows shipment activity by customer state and warehouse state. We created a State to WarehouseName hierarchy for drill-down. We also made a Drill-through Details page with fields such as ShipmentID, ServiceLevel, ShipmentType, DelayDays, OnTimeFlag and DamageFlag. This lets a user start from a summary chart and then look at the individual shipment records.

The dashboard is interactive, not just a group of fixed charts. Users can apply filters and slicers to focus on particular data. Drill-down shows more detail inside a hierarchy, while drill-through opens the related shipment records. The navigation buttons also make it easier to move around the report. This means a manager can quickly see the overall result and then check more detail when something looks unusual.

6. Business Recommendations

6.1 Improve Same Day reliability

Same Day had the lowest on-time rate at 83.53%, while Express was 87.64%. Same Day also had the highest median total cost in the service-level comparison. Since customers use this service for faster delivery, the company should look more closely at its performance. It could review dispatch cut-off times, route planning, carrier performance and vehicle availability. Tracking the Same Day on-time rate each week would also help show whether any changes are working.

6.2 Review Cold Chain delay

Cold Chain had the highest average delay at about 0.74 days. Management could check the parts of the process that are specific to Cold Chain, including special handling, loading time, temperature requirements and vehicle availability. These controls are necessary, so the aim is not to remove them. The main point is

to find where unnecessary waiting may be happening. Cold Chain still had a fairly strong on-time rate of 87.44%, so a focused review may be enough instead of changing the whole process.

6.3 Use distance when reviewing pricing and routes

Distance and TotalCost had a correlation of 0.738, so distance is clearly related to shipment cost. The company can use distance as one factor when checking prices and route efficiency. If a long-distance shipment has an unusually high cost, staff can check fuel use, tolls, carrier charges and the route used. Distance does not explain every cost, but it is a useful place to start.

6.4 Investigate expensive International and Same Day shipments

International shipments had an average cost that was much higher than the median. This happened because some International shipments were very expensive. Same Day also had the highest median cost. Management could look at the most expensive shipments in these two groups and check what caused the cost, such as distance, freight charges, fuel surcharge, insurance or special handling. The comparison should be done using one currency. If the company later wants a single financial view, it will need a proper currency-conversion method and a clear exchange-rate date.

6.5 Protect the high-volume Standard service

Standard handled 11,973 shipments, which was almost half of the cleaned shipment data. Its on-time rate was around 87.05%. Because Standard has such a large volume, a small drop in performance could affect many customers. The company should continue checking delivery time, damage rate and capacity for this service. Repeated problems should be investigated early. In the same way, even a small improvement in Standard can have a noticeable effect on the overall operation.

6.6 Reduce complaints and improve data quality

Deliveries with complaints had an average satisfaction score of 3.40, compared with 4.14 when there was no complaint. The company should record complaint reasons in a consistent way and check the common reasons each month. The problems we found in the original files also show that data entry needs better checking. Using fixed category lists, date validation, required ID checks and reasonable limits for numeric values could reduce data problems in the future and make the reports more reliable.

7. Conclusion

This project took the logistics data from its original raw form to a cleaned dataset and a Power BI dashboard. We used Python to find and fix data-quality problems, check the cleaned tables and create the first set of charts. After that, we used Power BI to connect the tables, calculate the KPIs and build interactive pages for shipment volume, delivery performance, cost, customers and warehouse operations.

The final results showed a few main areas to focus on. Overall on-time delivery was 86.93%, the average delay was 0.59 days and the damage rate was 3.03%. Same Day had the lowest on-time result, while Cold Chain had the highest average delay by shipment type. Complaint cases had lower customer satisfaction, and distance had a strong positive relationship with TotalCost. Standard had the highest shipment volume. Based on these findings, the company can focus on delivery reliability, route and cost checks, complaint handling and better data-quality controls.

Appendix A. Main DAX Measures

```
Total Shipments = COUNTROWS(Shipments_Clean)

On Time Rate = DIVIDE(CALCULATE(COUNTROWS(Shipments_Clean), Shipments_Clean[OnTimeFlag] =
"Y"), [Total Shipments], 0)

Average Delay = AVERAGE(Shipments_Clean[DelayDays])

Damage Rate = DIVIDE(CALCULATE(COUNTROWS(Shipments_Clean), Shipments_Clean[DamageFlag] =
"Y"), [Total Shipments], 0)
```